AirMesh

Crowdsourced, token-rewarded air quality sensing powered by Solana



The Problem & Our Solution

Cities lack affordable, granular air quality data; AirMesh crowdsources it via phones and cheap sensors, rewarded on Solana.

- Centralized sensor networks are too costly to deploy and maintain at city scale
- Air quality data is often sparse, outdated, or limited to city-wide averages
- AirMesh lets anyone submit geotagged readings from phones or cheap sensors
- Contributors earn SPL token rewards validated by on-chain consensus
- Result: hyperlocal, real-time air quality data for citizens and cities

How It Works

A mobile app captures signed sensor data, Solana verifies consensus, and rewards flow automatically to honest contributors.

- User flow: open app, take a reading, sign and submit geotagged data
- Readings stored via Arweave/IPFS, referenced on-chain for verification
- Switchboard oracle and multi-source consensus scoring validate accuracy
- Anchor program on Solana distributes SPL token rewards to contributors
- Staking mechanism discourages spam or fabricated submissions

Built with

Solana, Anchor, React Native, Mapbox, Switchboard oracle, Arweave/IPFS storage, SPL Token

Why Now & What's Next

DePIN is scaling fast on Solana's low fees, and demand for climate data has never been higher.

- Solana's low fees enable the micro-reward loops needed for DePIN networks
- Climate and environmental data demand is at an all-time high
- Next: partner with low-cost hardware sensor manufacturers for calibration
- Expand to new data types like noise and traffic congestion
- Launch a governance token and DAO for network parameter control

Category

DePIN